Isaac Pincus

Lausanne, Switzerland • ipincus@unil.ch • +41 77 943 38 54
Google Scholar • orcid.org/0000-0002-3826-9895

EDUCATION

Monash University, Melbourne, Australia Feb 2018 – Sep 2022
Doctor of Philosophy, Chemical Engineering
Research focused on simulating the rheology of polymer solutions, supervised by Professor Ravi Jagadeeshan (J. R. Prakash) at Monash University and Professor Alison Rodger at Macquarie University. The project examined the use of Brownian dynamics simulations to predict the linear dichroism of macromolecules and the rheology of dilute polymer solutions.

University of Queensland, Brisbane, Australia 2013 – 2017
Bachelor of Engineering (Hons.) / Bachelor of Science
Dual degree, majoring in Chemical Engineering and Physics.

RESEARCH EXPERIENCE

Postdoctoral Researcher — University of Lausanne 2024 – Present
Environmental Fluid Mechanics Group (Prof. Pietro de Anna)
Studying transport phenomena in unsaturated porous media.

Postdoctoral Associate — Massachusetts Institute of Technology 2022 – 2024
Qi Group and Non-Newtonian Fluids Group (Prof. Gareth H. McKinley)
Studied cellular biomechanics, fluid mechanics, and the rheology of polymer solutions.

Engineering Thesis Project — University of Queensland 2016 – 2017
Computational thesis project
Studied micelle formation using molecular dynamics simulations.

Rheology Research Project — University of Queensland 2017
Stokes Rheology Laboratory
Rheological research on a pharmaceutical project, working closely with scientists from a medical research centre.

Summer Research Student — University of Queensland 2014 – 2015
Rheology, Tribology and Biointerfaces Laboratory (Prof. Jason Stokes)
Sample preparation, data collection and analysis, and training in the use of rheometers.

MENTORING AND SUPERVISION

Co-supervisor

- **Fateme Sajedi** — PhD student, University of Lausanne, 2024–present.
- **Michele Caola** — PhD student, University of Lausanne, 2025–present.

Mentoring

- **Nadine Harmsen** — PhD student, University of Lausanne, 2024–present.
- **Bob Zhang** — graduate student, Massachusetts Institute of Technology, 2022–2024.
- **Nicholas King** — graduate student, Massachusetts Institute of Technology, 2022–2024.
- **Samuel Coutts** — undergraduate summer student, Massachusetts Institute of Technology, 2023.

TEACHING

University of Lausanne

- Vector Analysis — **Lecturer**, 2025 and 2026.
- Advanced fluid dynamics — informal course developed and taught to graduate students in the department.

Massachusetts Institute of Technology

- Advanced Transport Phenomena — assisted with teaching in an unofficial capacity, 2022–2024.

Monash University *(Teaching Associate, 2018–2021)*

- CHE2161 Mechanics of Fluids — demonstrator.
- CHE3167 Transport Phenomena and Numerical Methods — demonstrator.
- CHE4162 Particle Technology — demonstrator.
- CHE6112 Advanced Fluid Dynamics — demonstrator.
- Ran tutorials and laboratory sessions and marked assignments and exams. Led a revamp of the fluid mechanics laboratory practicals and designed instructional videos for the transport phenomena classes.

University of Queensland

- CHEE2003 Fluid and Particle Mechanics — tutor.
- PHYS1001 Mechanics & Thermal Physics 1 — tutor.

Teaching materials

- Developed a suite of MATLAB scripts with dynamic figures for visualising vector fields in 3D, used in teaching vector analysis. Available at isaacpincus.github.io/teaching.

PUBLICATIONS

Peer-reviewed journal articles

1. **Pincus, I.**, & Qi, Q. M. (2025). Nanoparticle-induced lipid membrane deformation influences the design of biomedicine. *Biophysical Journal*. doi:10.1016/j.bpj.2025.10.030.
2. **Pincus, I.**, Rodger, A., & Prakash, J. R. (2024). Flow dichroism of DNA can be quantitatively predicted via coarse-grained molecular simulations. *Biophysical Journal*, 123(21), 3771–3779.
3. **Pincus, I.**, Rodger, A., & Prakash, J. R. (2023). Dilute polymer solutions under shear flow: comprehensive qualitative analysis using a bead-spring chain model with a FENE-Fraenkel spring. *Journal of Rheology*, 67(2), 373–402.
4. **Pincus, I.**, Rodger, A., & Prakash, J. R. (2020). Viscometric functions and rheo-optical properties of dilute polymer solutions: comparison of FENE-Fraenkel dumbbells with rodlike models. *Journal of Non-Newtonian Fluid Mechanics*, 285, 104395.

Preprints and manuscripts under review

5. **Pincus, I.**, Calabrese, V., Haward, S. J., & McKinley, G. H. (2026). Numerical analysis of capillarity-driven thinning rheometry for polydisperse polymer solutions. arXiv:2607.10376.
6. Dong, J., Kim, J. H., **Pincus, I.**, Manna, S., Podgorski, J. M., Zhu, Y., & Tadesse, L. F. (2025). Interplay of electrostatic interaction and steric repulsion between bacteria and gold surface influences Raman enhancement. arXiv:2501.06759.
7. Burey, P., Shewan, H. M., Bonilla, M. R., **Pincus, I.**, Shelat, K., Yakubov, G., & Stokes, J. R. Predicting thin-film rheological behaviour as a function of particle micromechanics using gap-dependent rheology (GDR). *Under review.*

PRESENTATIONS AND SEMINARS

- **Pincus, I.**, Harmsen, N., Quet, J., de Anna, P., "Modelling biofilm growth and detachment in porous media," **Invited talk**, BugControl Workshop, Utrecht, Netherlands, 2026.
- **Pincus, I.**, Harmsen, N., Quet, J., de Anna, P., "Modelling biofilm growth and drying in 2D porous media," **Invited talk**, Ghent University, Belgium, 2026.
- **Pincus, I.**, Harmsen, N., Sajedi, F., Caola, M., Quet, J., de Anna, P., "Drying, transport and biofilm growth in heterogeneous porous media," Non-Newtonian Fluid Mechanics in Porous Media Conference, Cargèse, France, 2026.
- **Pincus, I.**, de Anna, P., "Drying in heterogeneous porous media," InterPore Annual Meeting, Nantes, France, 2026.
- **Pincus, I.**, Calabrese, V., Haward, S. J., McKinley, G. H., "Capillary breakup extensional rheometry of polydisperse polymer solutions," Annual European Rheology Conference (AERC), Lyon, France, 2025.
- **Pincus, I.**, Rodger, A., Prakash, J. R., "Rheology and linear dichroism of semiflexible polymers in shear flow via Brownian dynamics," Structure, Stress, Strain & Stretching Flows (S4F), CECAM Workshop, Monash University Prato Centre, Prato, Italy, 2025.
- Yamani, S., **Pincus, I.**, Prakash, J. R., McKinley, G. H., "Accurately capturing shear and extensional rheology of xanthan gum: a FENE-Fraenkel constitutive equation with distinct orientation and stretch timescales," Society of Rheology Annual Meeting, Austin, United States, 2024.
- **Pincus, I.**, Rodger, A., Prakash, J. R., "Rheology and linear dichroism of semiflexible polymers in shear flow via Brownian dynamics," Non-Newtonian Flows in Porous Media, Banff International Research Station, Canada, 2024.
- **Pincus, I.**, Qi, Q. M., "Modelling the effects of particle surface loading on cell deformation," Poster presentation, International Congress on Rheology, Athens, Greece, 2023.
- **Pincus, I.**, Rodger, A., Prakash, J. R., "Rheology and linear dichroism of semiflexible polymers in shear flow," Invited talk, Future of Rheology Seminar Series, 2023.
- **Pincus, I.**, Rodger, A., Prakash, J. R., "Rheology and linear dichroism of semiflexible polymers in shear flow," Poster presentation, Society of Rheology Annual Meeting, Chicago, United States, 2022.
- **Pincus, I.**, Rodger, A., Prakash, J. R., "Brownian dynamics for polymer rheology," Seminar, Monash HPC Seminar Series, Melbourne, Australia, 2022.
- **Pincus, I.**, Rodger, A., Prakash, J. R., "Use of a novel spring potential in rheology of dilute polymer solutions," Presentation, International Congress on Rheology, Rio de Janeiro, Brazil (remote), 2020.
- **Pincus, I.**, Prakash, J. R., "FENE-Fraenkel dumbbell rheology and statistical mechanics," Poster presentation, Statistical Mechanics of Soft Matter Meeting, Adelaide, Australia, 2019.

AWARDS AND HONOURS

- Jim Swan Graduate Student Poster Award, International Congress on Rheology, 2023.
- Society of Rheology Student Travel Grant, SoR Annual Meeting, Chicago, 2022.
- Society of Rheology Student Travel Grant, International Congress on Rheology, Rio de Janeiro, 2020 (not taken up — conference held online).
- Australian Government Research Training Program Scholarship, 2018–2022.

PROFESSIONAL SERVICE

Peer review

- *Macromolecules* (ACS)
- *Energies* (MDPI)
- *Journal of Geophysical Research: Solid Earth* (AGU)
- *Journal of Hydrology* (Elsevier)
- *Journal of Polymer Science* (Wiley)
- *Journal of Fluid Mechanics* (Cambridge University Press).

Service and outreach

- MIT Chemical Engineering Postdoctoral Advisory Board, 2022–2024. Organised bimonthly research seminars for postdoctoral scholars in MIT Chemical Engineering, alongside regular social events.
- Academic Vice-President, Chemical Engineering Postgraduate Association, Monash University, 2019–2020. Central role in creating and running the CEPA 2019 student conference (100+ attendees from research, academia, and industry), which has run as an annual event since; minor role in the Monash Engineering 2021 student conference.
- Student representative, Monash University Graduate Research Committee, 2020. Monthly meetings with Faculty and University representatives on behalf of the graduate student body.
- Volunteer high-school debating coach, 2013–2015.

TECHNICAL SKILLS

- **Research areas:** rheology and soft-matter physics, particularly polymer solution rheology; transport, drying, and biofilm growth in porous media; cellular and membrane biomechanics.
- **Simulation:** Brownian dynamics; coarse-grained molecular simulation; molecular dynamics; continuum and pore-scale flow solvers (OpenFOAM).
- **Programming:** MATLAB and Fortran (advanced); Python, C, C++, and Java (working knowledge); high-performance and parallel computing.
- **Microfluidics and fabrication:** design and fabrication of microfluidic devices; soft lithography; PDMS moulding and casting; NOA-81 (UV-curable adhesive) device printing and bonding; associated microscopy and time-lapse imaging.
- **Rheometry:** rotational and extensional rheometry; rheo-optical methods (linear dichroism).

SOFTWARE

- `BiofilmGrowthFoam` (OpenFOAM) — solver suite for biofilm growth in porous media, including a custom library for spatially varying Herschel–Bulkley rheology.
- `polydisperseCapillaryBreakup` (MATLAB) — multi-mode FENE-PM model of capillarity-driven filament thinning for polydisperse solutions; every figure in Pincus et al. (2026) can be regenerated from the included scripts.
- `SingleChainBD` (Fortran) — Brownian dynamics simulation of single polymer chains; underpins the bead-spring chain results in Pincus et al., *J. Rheol.* (2023).
- `General_Dumbbell` (Fortran) — general dumbbell simulator supporting multiple spring laws and integrators; used in Pincus et al., *JNNFM* (2020).
- `MembraneStretching` (MATLAB) — particle-membrane interaction and membrane deformation model from Pincus & Qi, *Biophys. J.* (2025).
- Full repository list at github.com/IsaacPincus.

INTERESTS

Social football, volleyball, rock climbing, windsurfing, reading novels.

REFERENCES

- **Professor Ravi Jagadeeshan (J. R. Prakash)** — Professor, Department of Chemical and Biological Engineering, Monash University.
  +61 3 9905 3274 • Ravi.Jagadeeshan@monash.edu
- **Professor Pietro de Anna** — Institute of Earth Sciences, University of Lausanne. Postdoctoral supervisor.
  +41 21 692 44 23 • Pietro.DeAnna@unil.ch
- **Professor Gareth H. McKinley** — Department of Mechanical Engineering, Massachusetts Institute of Technology. Postdoctoral supervisor.
  +1 617 258 0754 • gareth@mit.edu
- **Professor Alison Rodger** — Director, Research School of Chemistry, Australian National University.
  +61 426 627 077 • alison.rodger@anu.edu.au
- **Professor Jason Stokes** — Professor, School of Chemical Engineering, University of Queensland, and Leader of the Premium Food and Beverages Program, Food and Beverage Accelerator (FaBA).
  +61 7 3365 4361 • jason.stokes@uq.edu.au